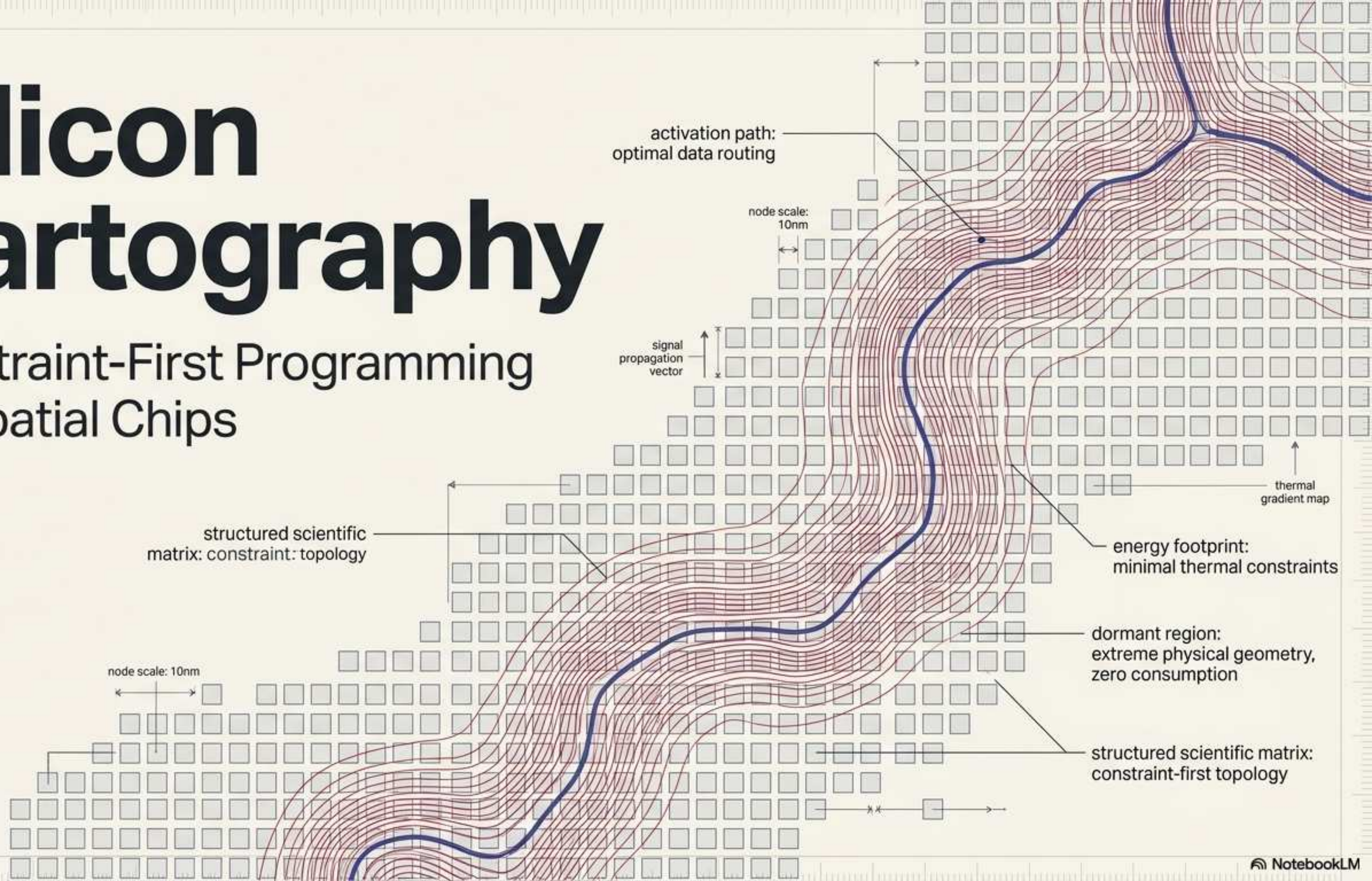


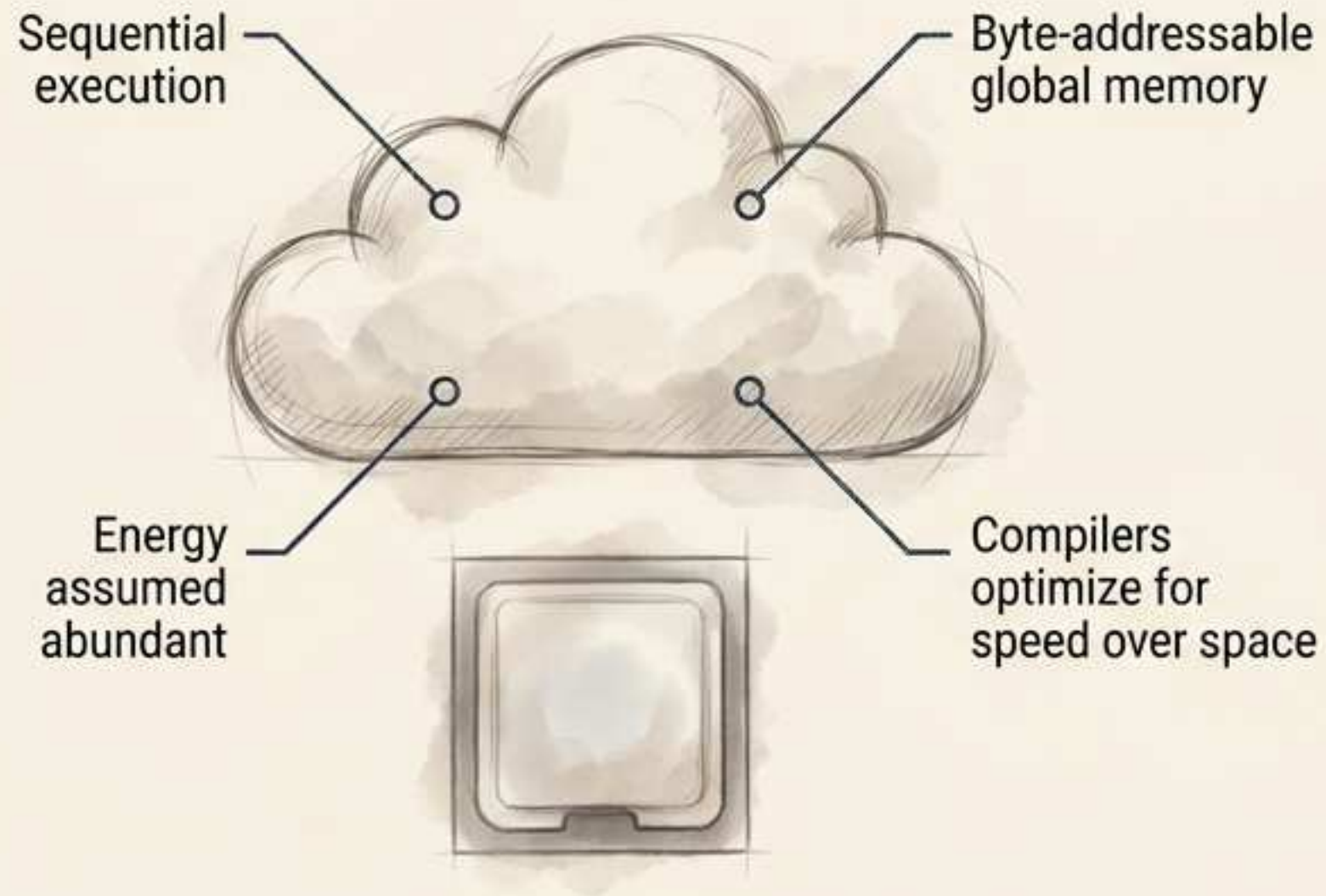
Silicon Cartography

Constraint-First Programming for Spatial Chips

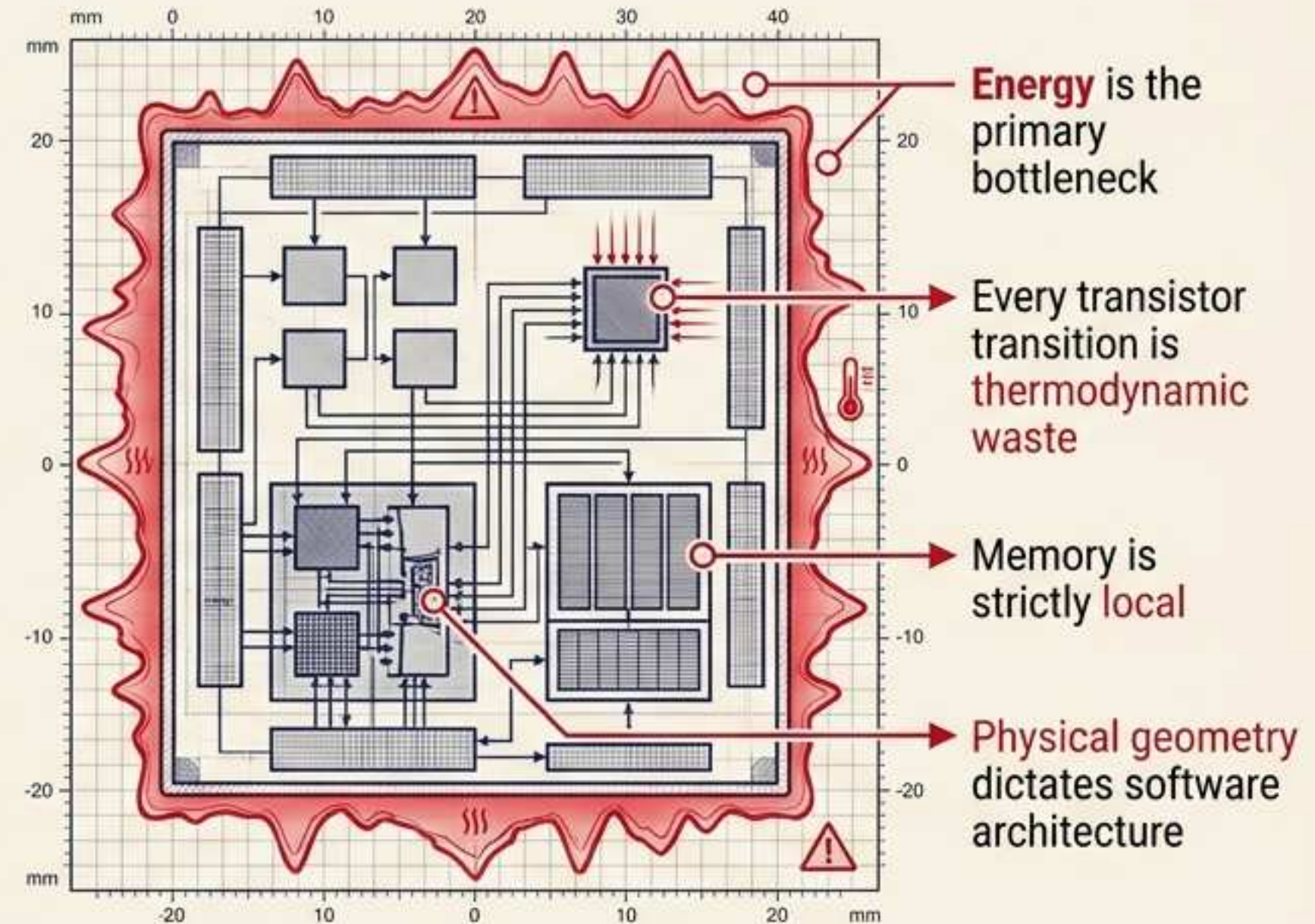


The End of the Uniform Abstraction Era

1970–2010: The Illusion of Infinite Resources

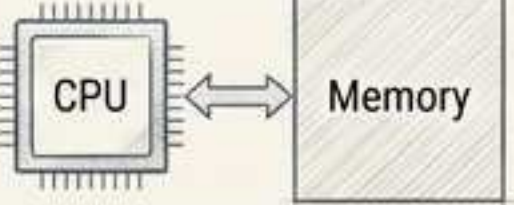
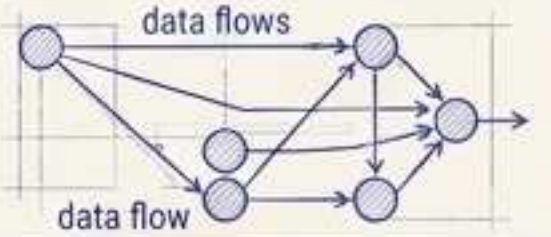
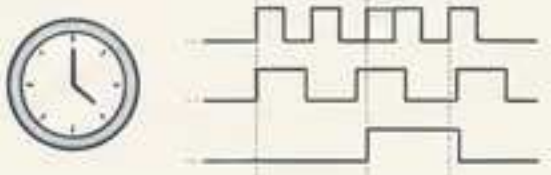
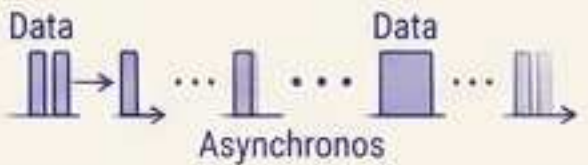
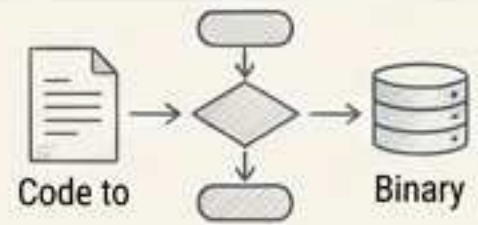
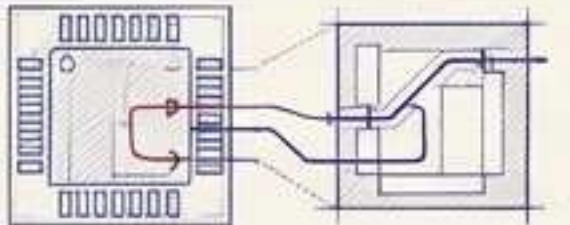


Present & Future: The Return of Physicality



Key Takeaway: At the spatial and temporal scales of modern embedded systems, optimization is no longer about rewriting syntax—it is about finding the most energy-minimal physical placement of computation.

From Sequential Control to Constraint Propagation

	The von Neumann Inheritance	Silicon Cartography
Architecture	Uniform, sequential, centralized mutable state. 	Spatial, metrized, distributed constraint graphs. 
Energy	Ignored as a secondary parameter. 	The primary limiting functional constraint. 
Execution	Clock-driven, global synchronization. 	Sparse asynchronous activation (event-driven). 
Compiler's Role	Semantic translator and speed optimizer. 	Spatial placer and routing constraint solver. 

Sequential models externalize coordination into scheduling overhead.
Constraint-propagation models internalize coordination into the physical admissibility structure itself.

The Cartographer's Methodology

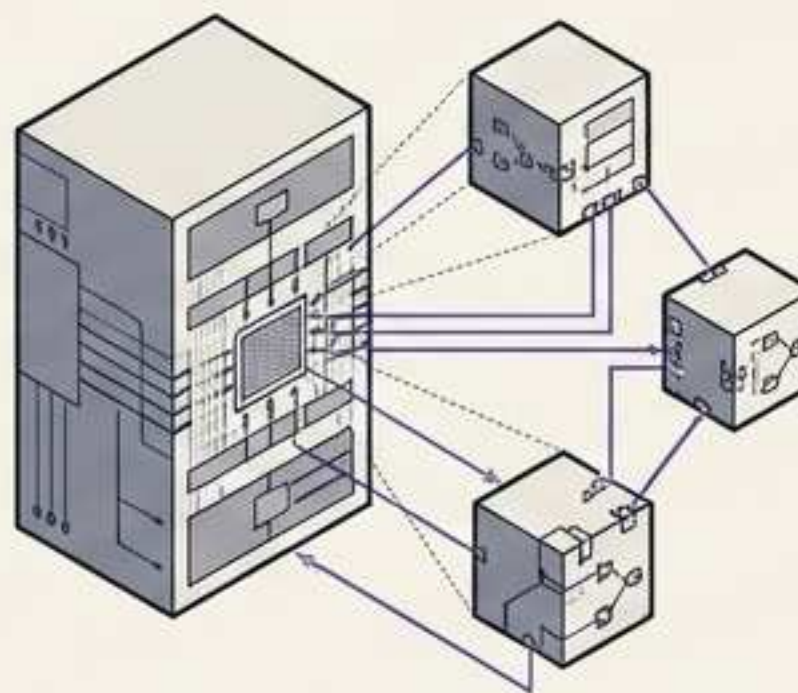
You Are Here

Interrogate the Hardware



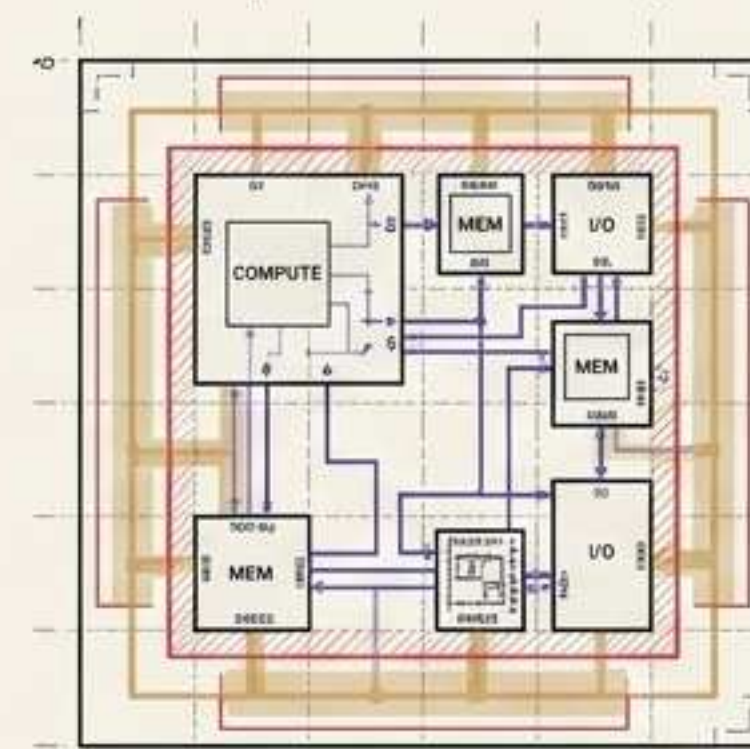
1 Extract the executable constraint map from the physical silicon before writing a line of code.

Decompose the Program



2 Factor computation into locally self-contained kernels with explicit message interfaces. No global memory.

Map to Geometry



3 Solve the spatial routing problem to minimize physical energy cost and preserve locality.

The Territory: Modeling the Metrized Constraint Graph

$$H = (N, E, I, M, C, P)$$

Nodes & Edges

N (Compute Cores) and **E** (Directed Communication Links).

Instruction & Memory

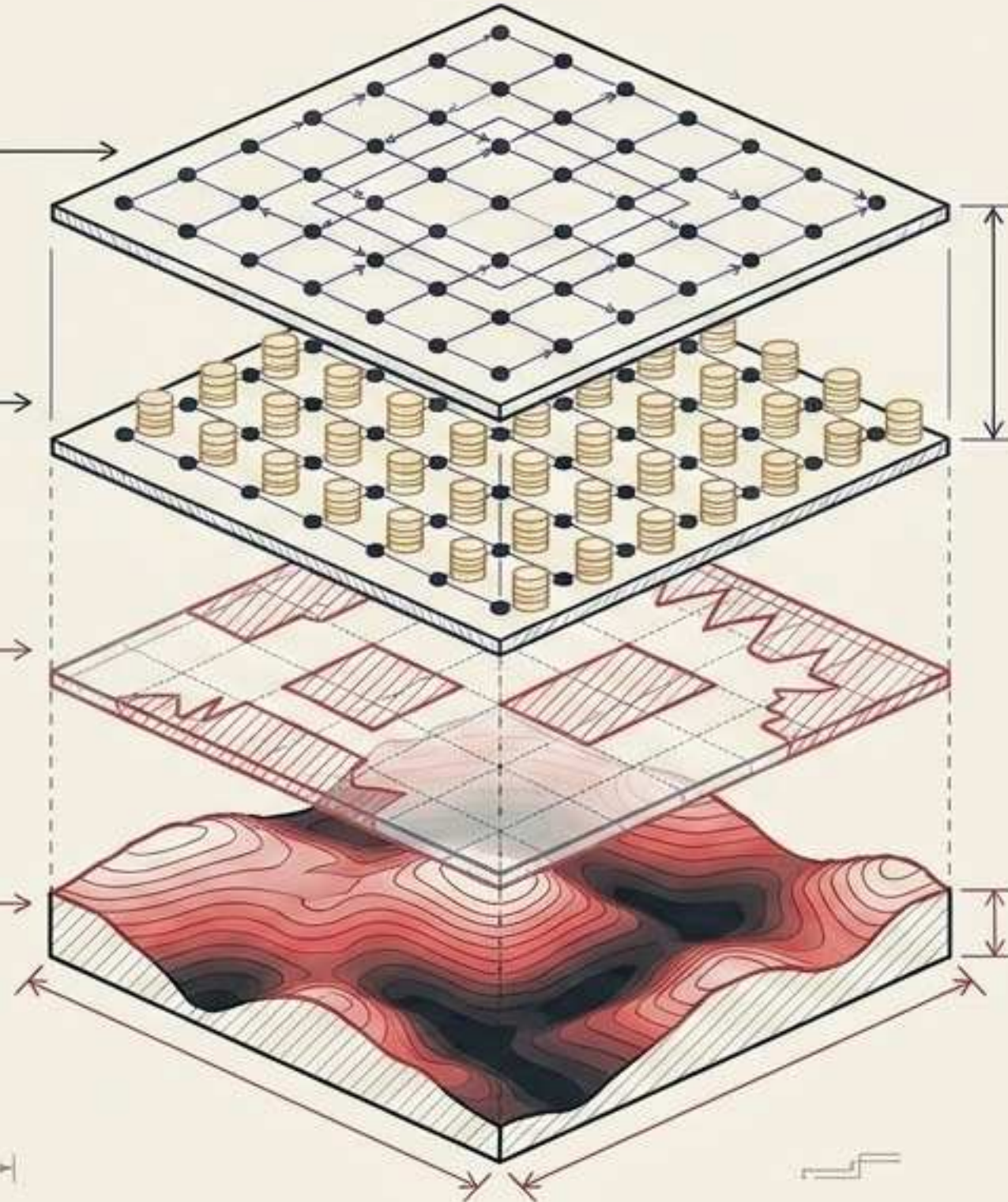
I (Local Instruction Algebras) and **M** (Strictly Local Memory Geometry).

Constraints

C (Hard Constraints: maximum fanout, routing exclusions, timing windows).

Energy Topology

P (Physical Cost Model: energy, latency, and heat of transitions).



The Expedition: Locally Self-Contained Computation

$$G = (K, D, R)$$

Program Graph

Visualizing K (Kernels)

Isolated computational units operating strictly on local data.

Visualizing D

Explicit message passing. Global address spaces do not exist.

Visualizing R (Requirements)

Resource requirements dictated before placement.

Visualizing D (Dependencies)

Explicit message passing. Global address spaces do not exist.

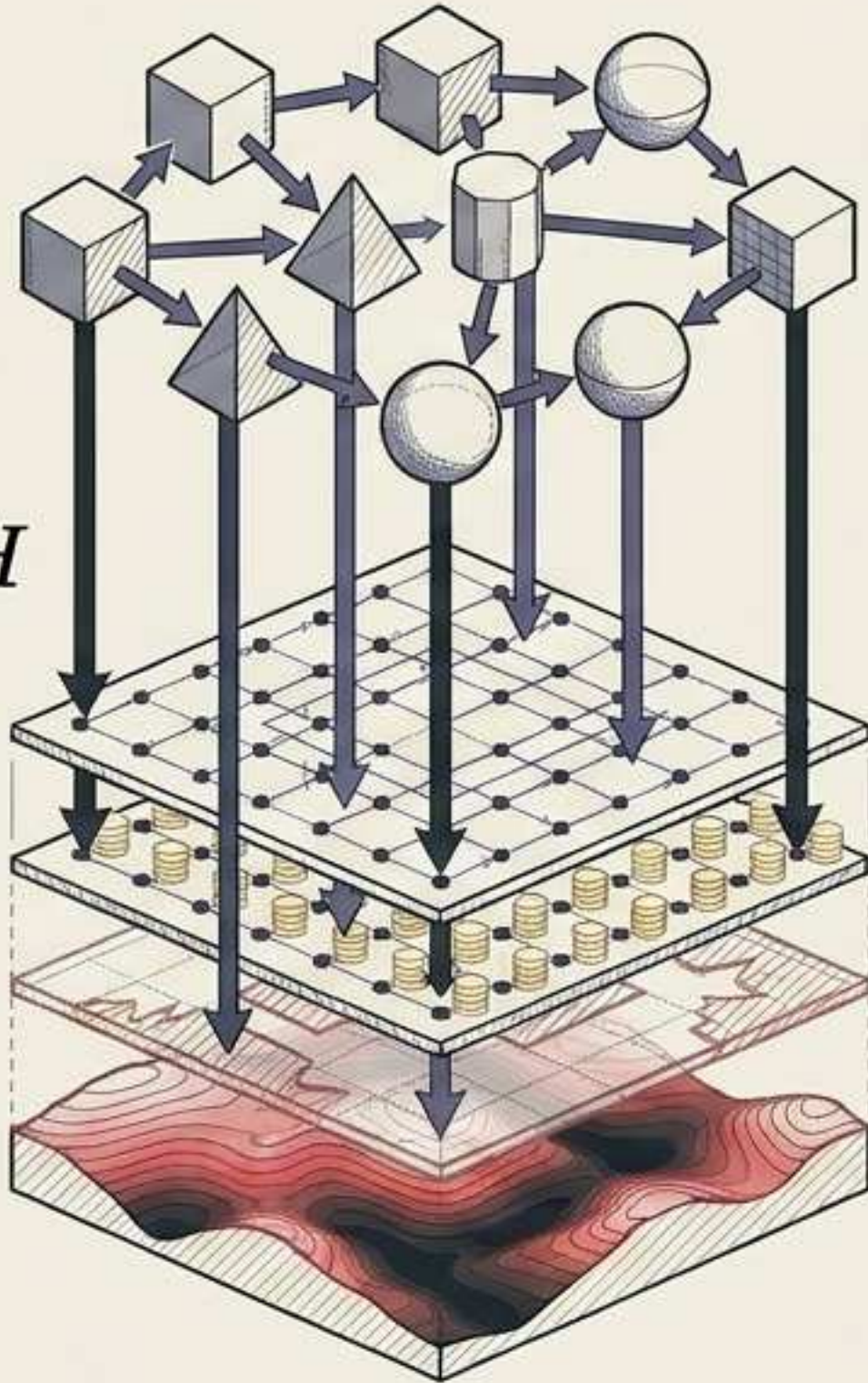
Key Takeaway: A well-written program on a spatial chip looks like a sparse activation field, not a dense, centralized computation trace.

The Mapmaker: Solving the Admissibility Field

Program
Graph

$\varphi : G \rightarrow H$

Hardware
Graph



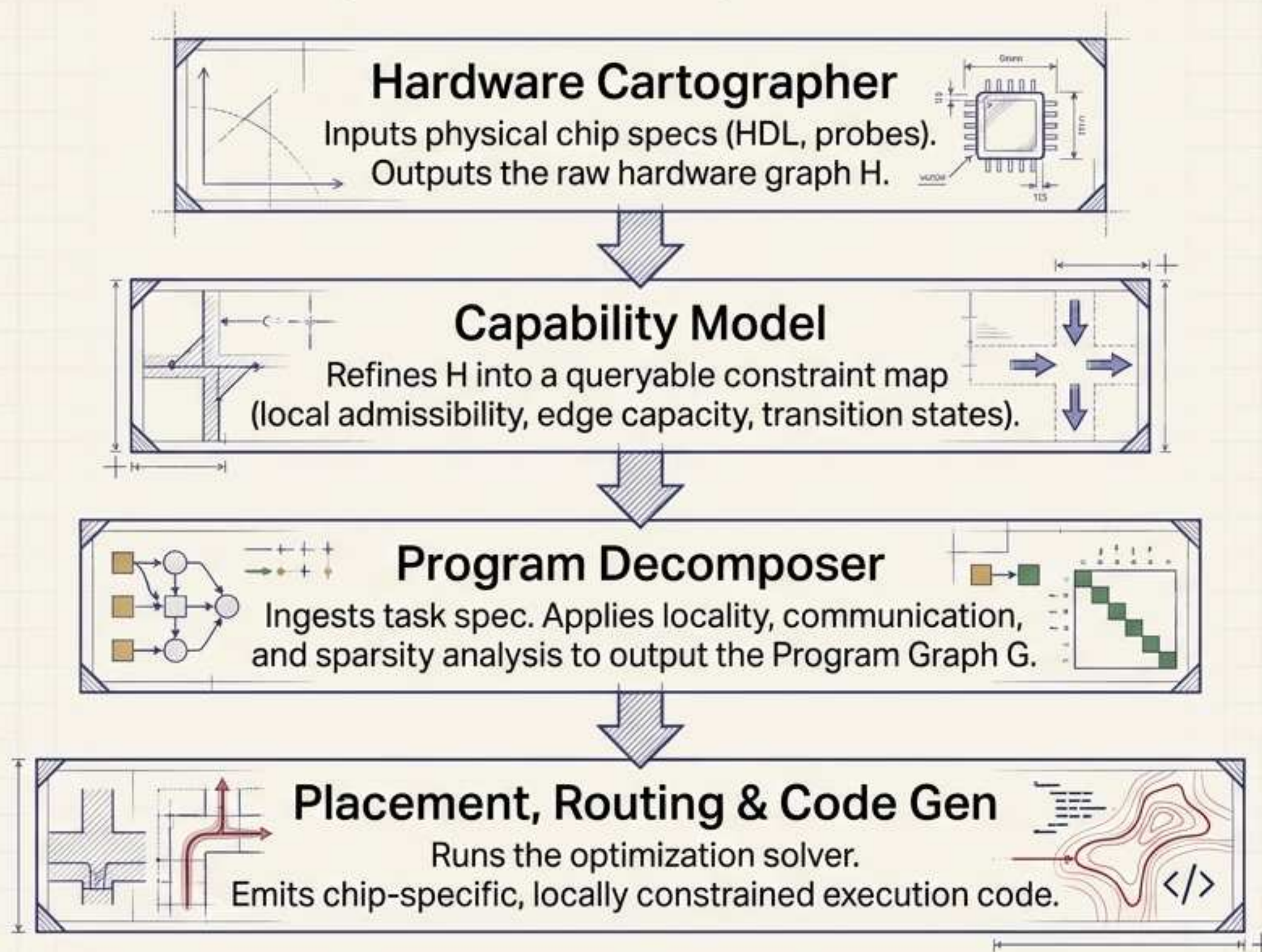
Optimization Equation

$$\begin{aligned} Cost(\varphi) = & \alpha E_{\text{exec}} + \beta E_{\text{comm}} \\ & + \gamma L_{\text{critical}} + \delta M_{\text{pressure}} + \epsilon S_{\text{waste}} \end{aligned}$$

αE_{exec}	Total execution energy
βE_{comm}	Total communication energy (routing cost)
$\gamma L_{\text{critical}}$	Critical-path latency
$\delta M_{\text{pressure}}$	Peak memory pressure
$\epsilon S_{\text{waste}}$	Synchronization waste (active waiting)

The compiler acts as a spatial constraint solver over hardware geometry, finding the mapping ϕ that minimizes thermodynamic cost.

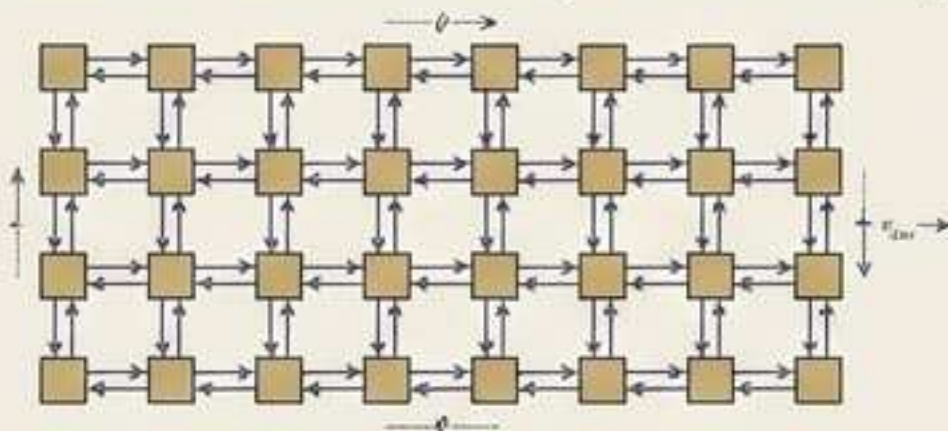
The Four-Layer Compiler Architecture



A Universal Cartography for Spatial Substrates

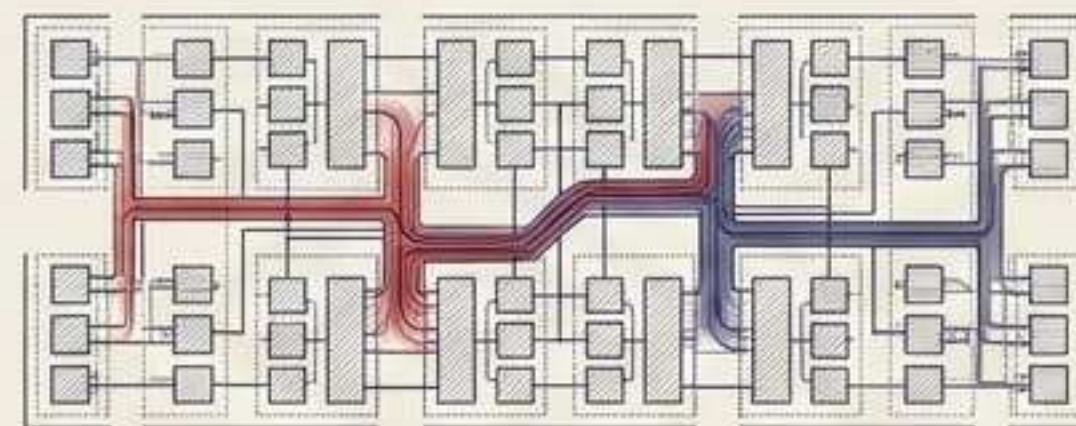
The framework generalizes across diverse spatial architectures.

Mesh Processors (GreenArrays)



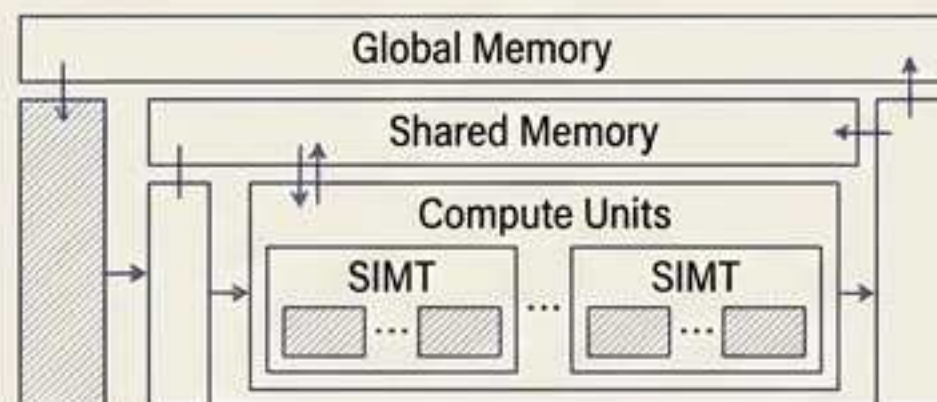
Blocking synchronous rendezvous.
Energy proportional to port wire transitions.

FPGAs



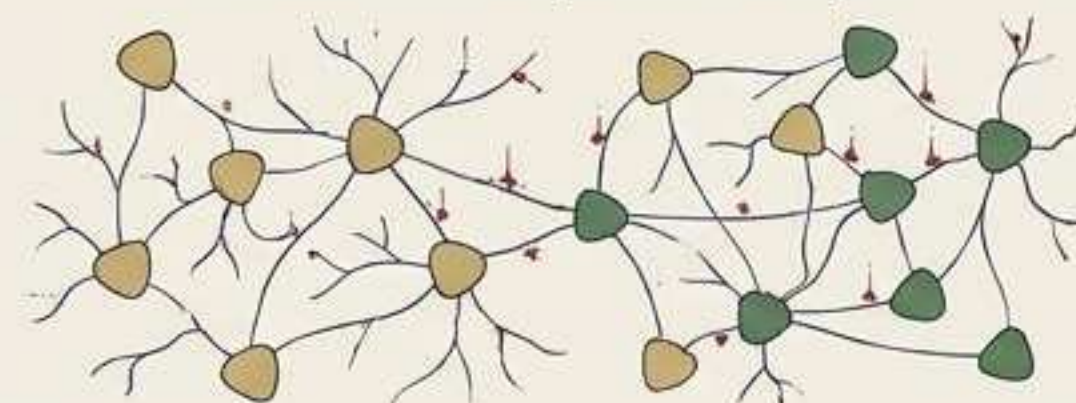
Configurable computation where routing interconnect
congestion is the primary bottleneck.

GPUs



Massive parallelism bound by strict memory
hierarchy and SIMT execution constraints.

Neuromorphic Chips

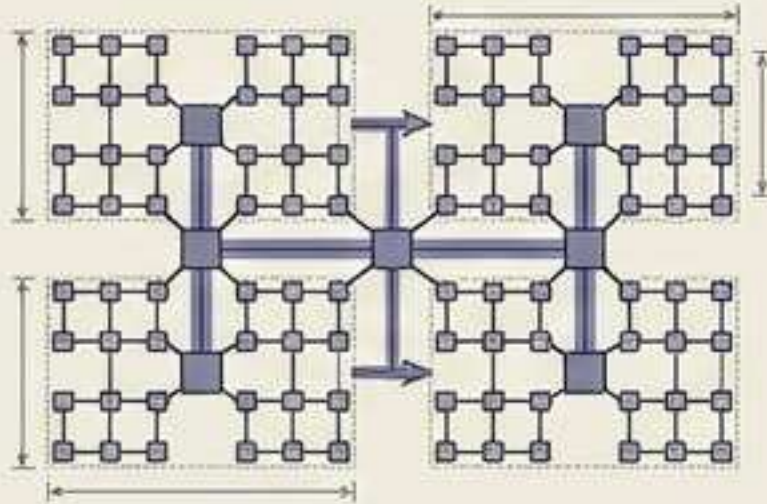


Asynchronous spike propagation optimized
for minimal activation events.

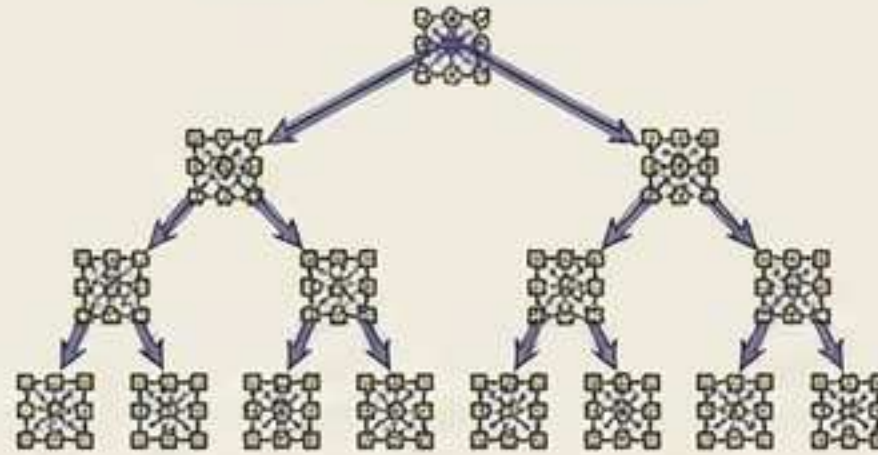
Fractal Topology Families

Moving beyond the uniform flat grid to design chips for specific communication localities.

Fractal Island Mesh

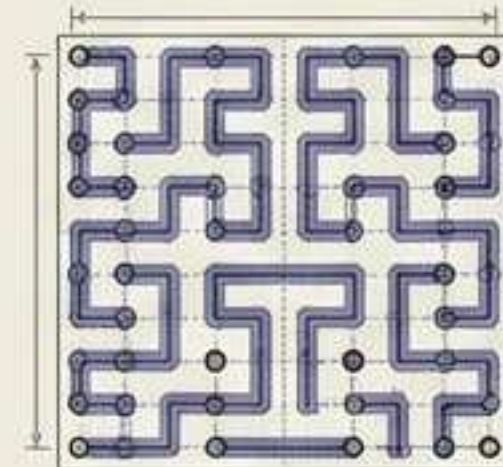


Tree-of-Meshes



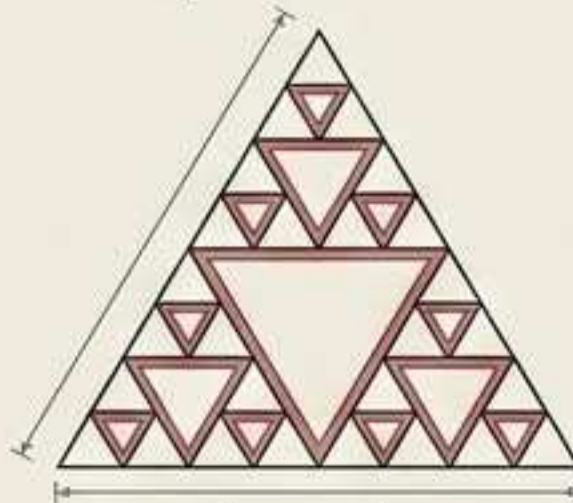
Ideal for sensor swarms.

Hilbert-Curve



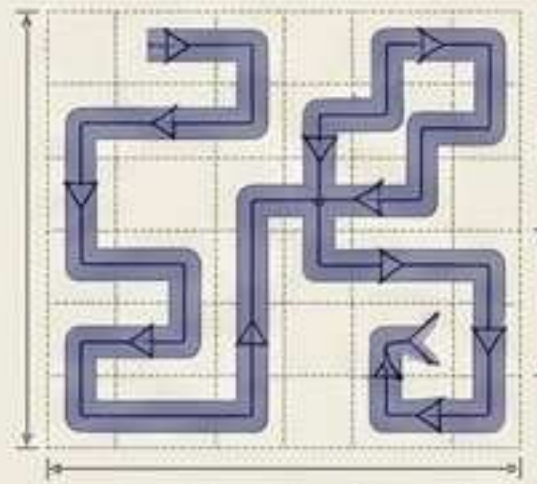
Transforms 1D streams into 2D locality.

Sierpinski Fabric



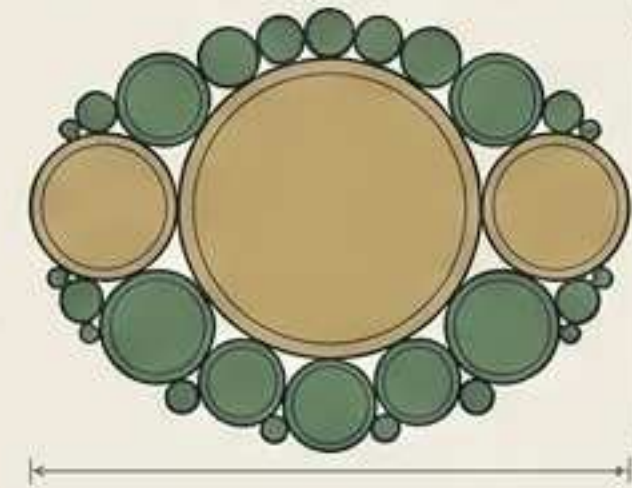
Voids serve as thermal separation gaps.

Dragon-Curve

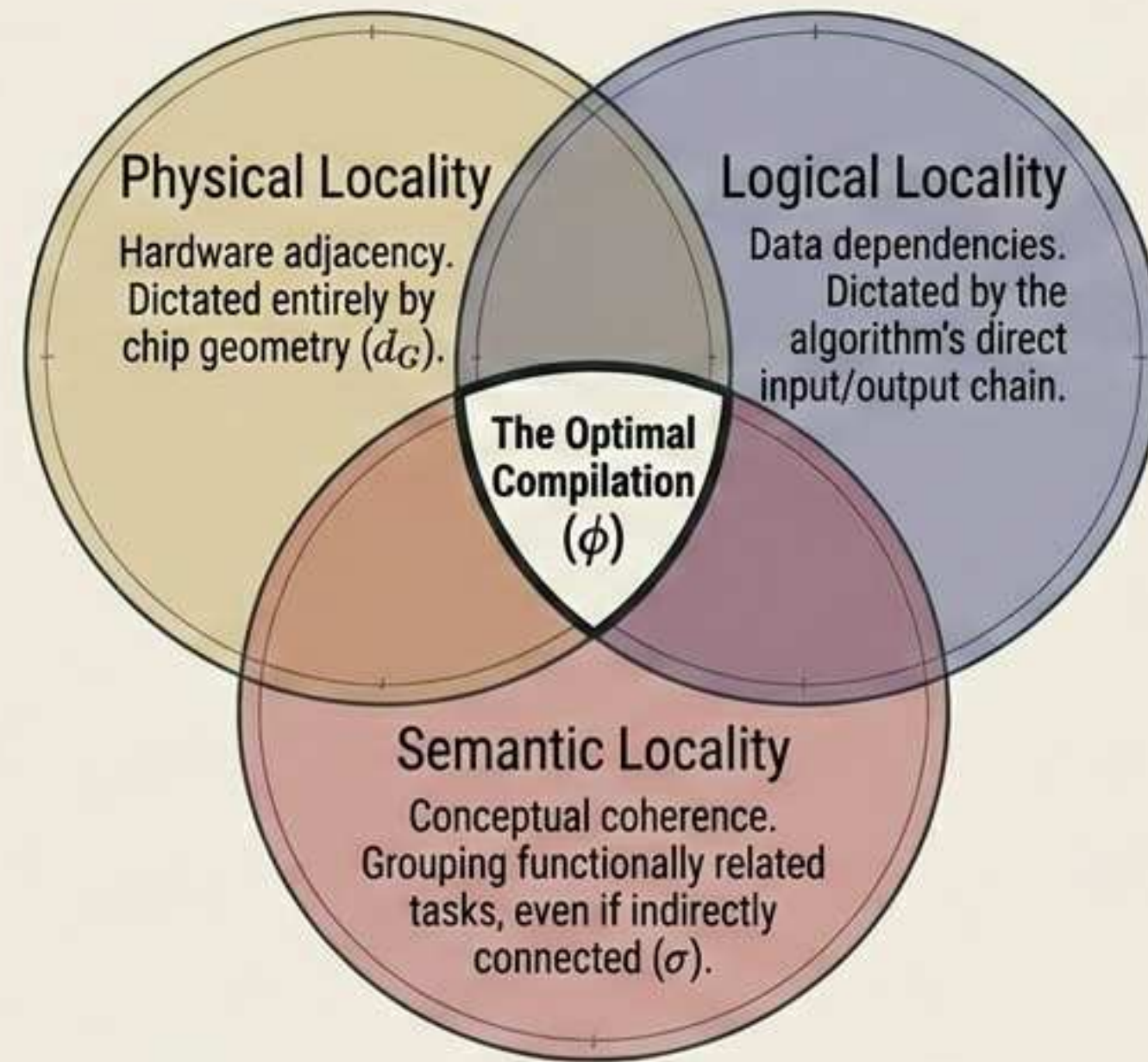


Asynchronous pipeline for localized streaming.

Apollonian Heterogeneous Fabric



The Three Senses of Locality



$$\text{The Lipschitz Condition: } d_G(\phi(k_i), \phi(k_j)) \leq L \|\sigma(k_i) - \sigma(k_j)\|$$

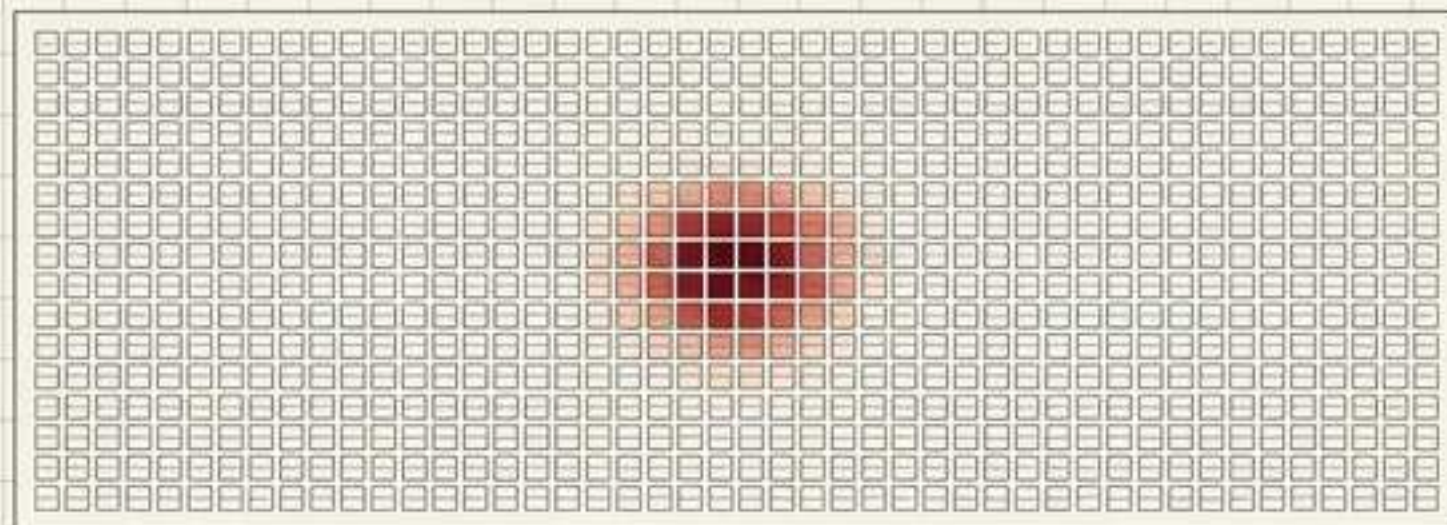
A perfect compilation embeds the algorithm's semantic geometry directly into the physical constraints of the silicon.



Energetic Cartography: Visualizing the Unseen

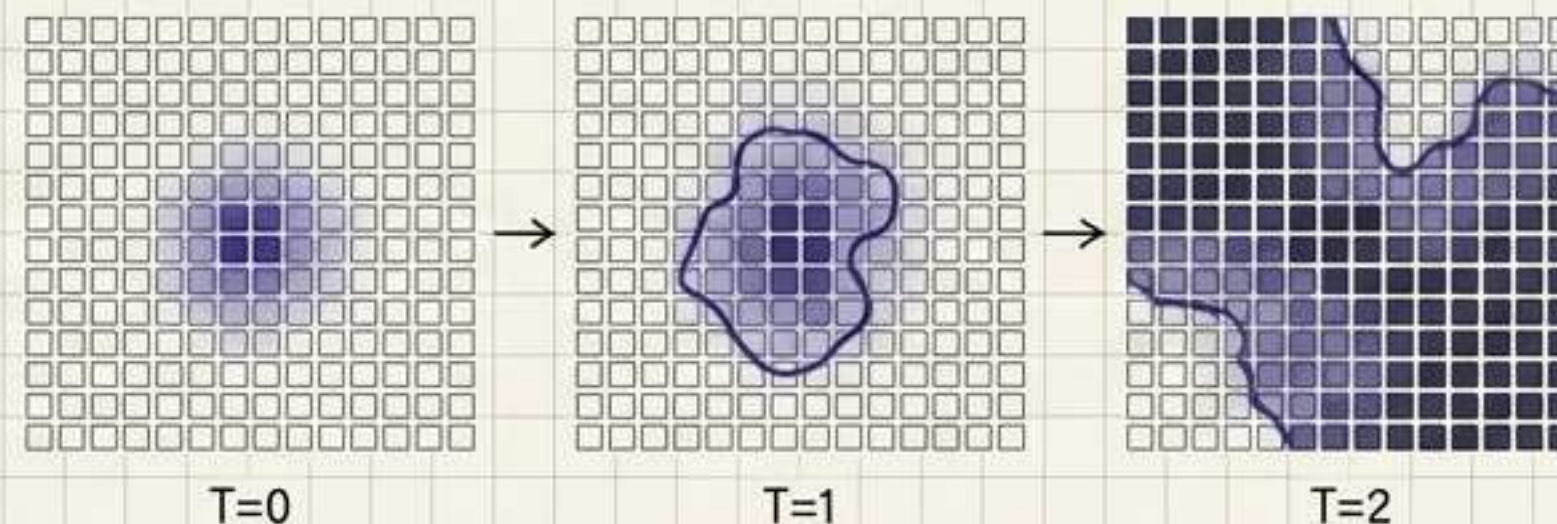
Block diagrams are inadequate. We must visualize the 6 dimensions of spatial chips.

Activity Heatmaps



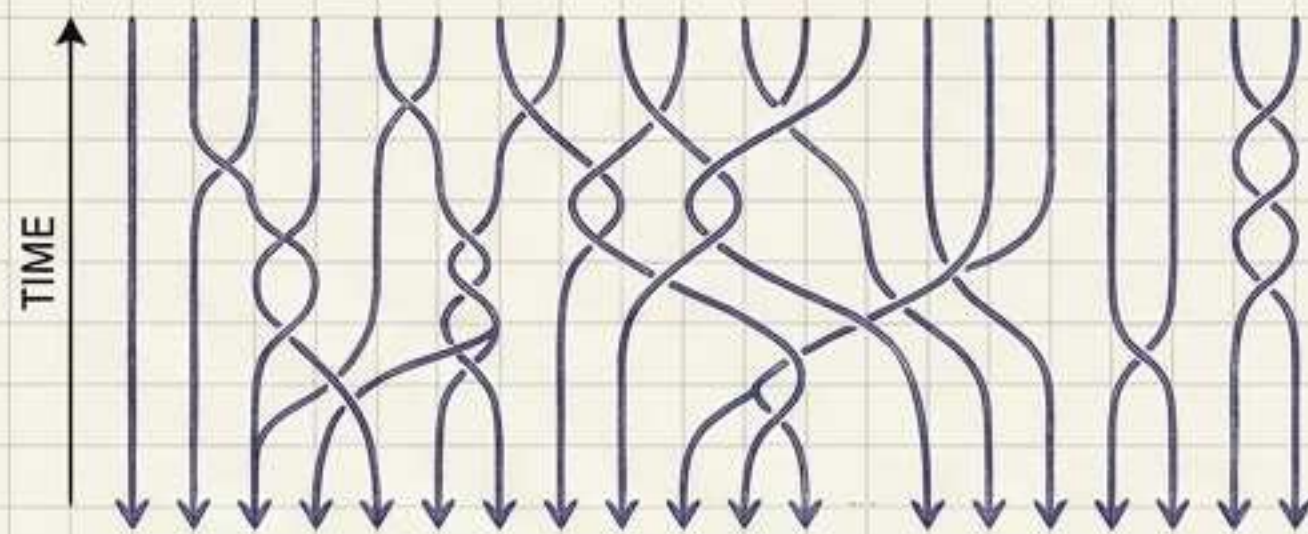
Darkness is success. High activity appears as localized warm colors.

Event Waves



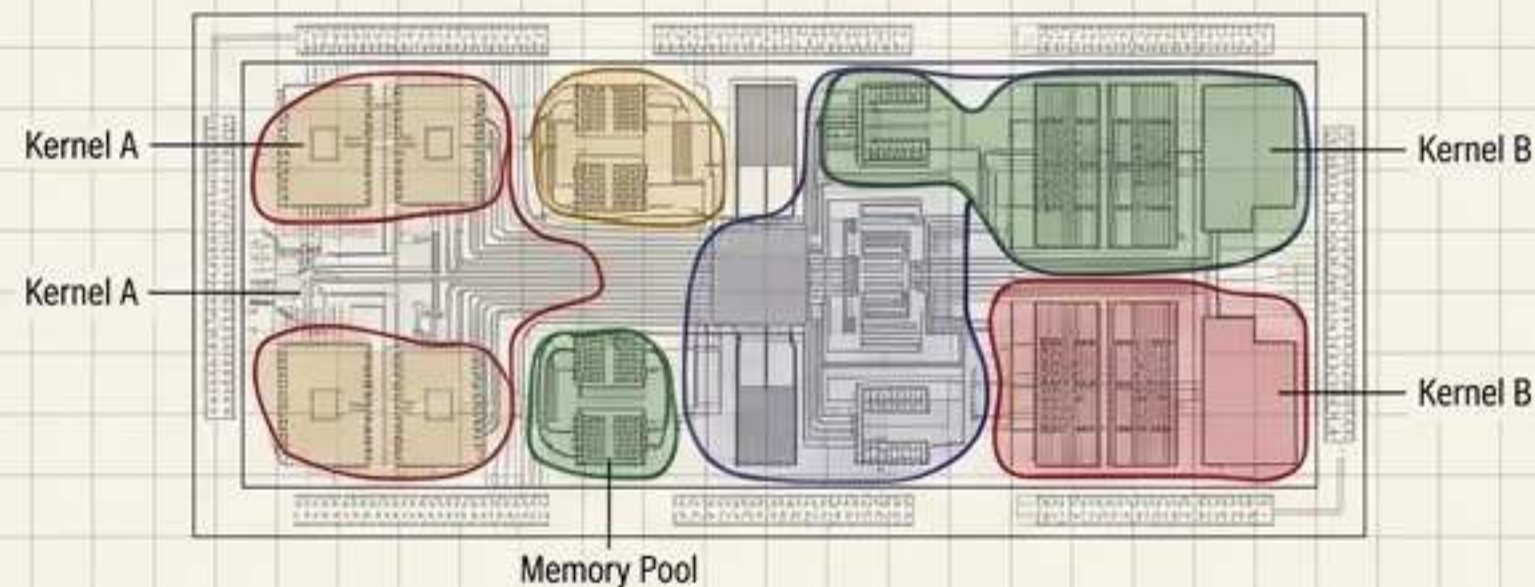
Pulses propagate organically, proving the absence of a global clock.

Causal Braid Diagrams



Replaces clock-cycle diagrams to expose synchronization deadlocks and causal dependencies.

Compiler Overlays

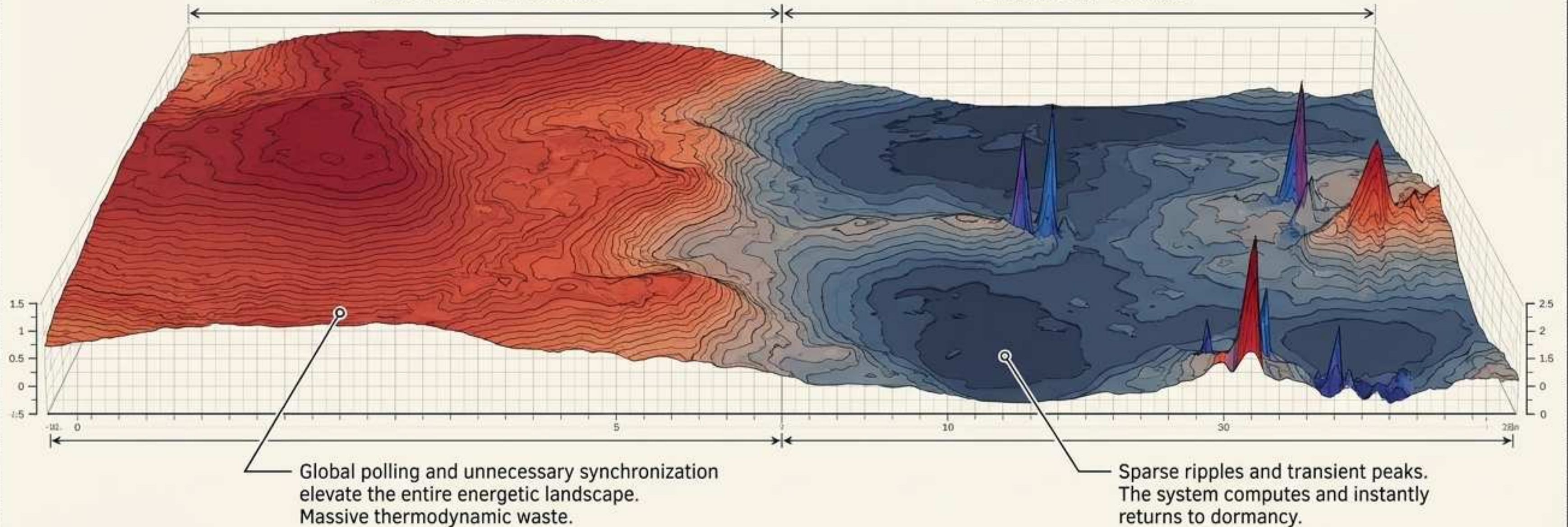


Maps software kernels to physical hardware real estate.

The Energy Landscape Terrain

Inefficient Code

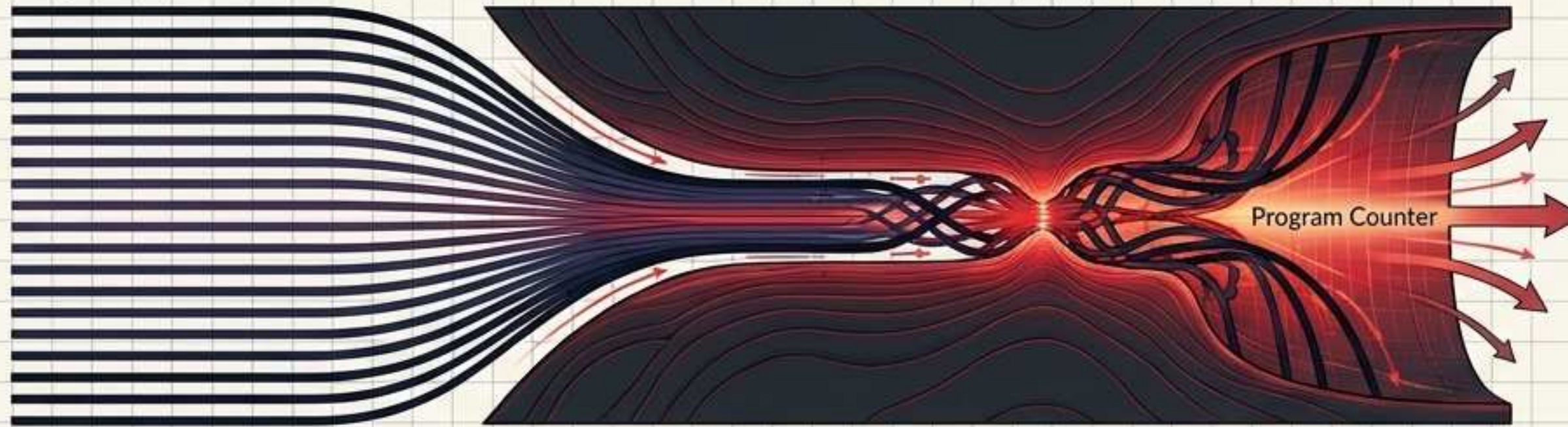
Efficient Code



Key Takeaway: Unnecessary switching is waste, visible as unnecessary physical height in the energetic geography.

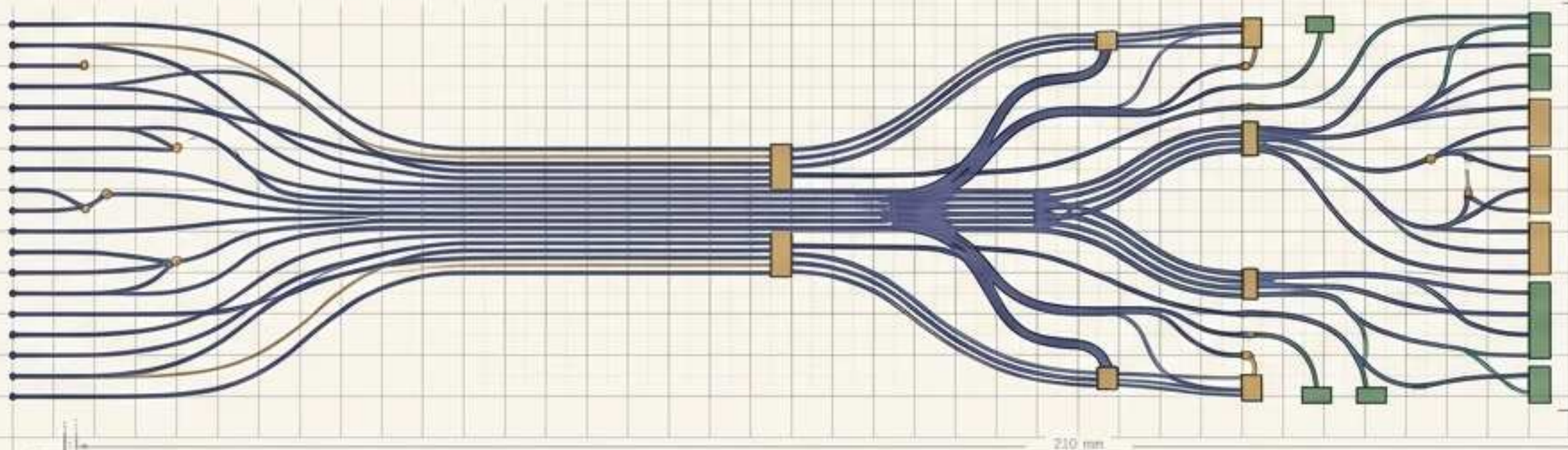
The Thermodynamic Cost of Serialization

Sequential Execution



Artificial Serialization:
Forcing distributed
independent data to
synchronize perfectly
requires massive
coordination energy.

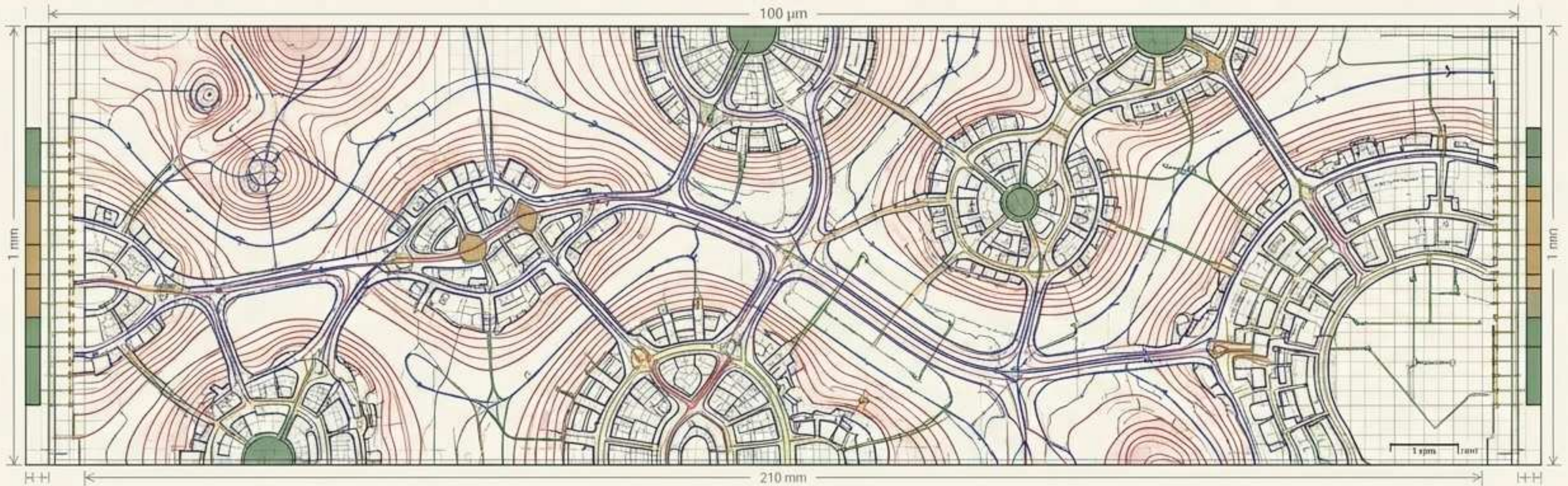
Sparse Asynchronous Activation



Spacetime Activity Volume:
Computation distributes
naturally without global
coordination overhead.

Landauer's principle proves that discarding information carries an inescapable thermodynamic price.
Serialization is an extreme, artificial restriction on the accessible manifold of low-energy computational states.

The Machine Ecology



From Commands to Constraints

Programming is no longer about dictating sequential steps, but designing local propagation rules.

From Abstraction to Reality

We must stop hiding the physical substrate and begin negotiating with the hardware geometry.

Computing Honestly

Treating energy minimality not as a final optimization pass, but as the foundational law of spatial architecture.

**Silicon Cartography
recovers the direct
negotiation between
algorithm and matter.**